

WASP-104b

R-2621

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150527 · created 2026-08-28T23:30:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457169.71257 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.044
Transit depth (Rp/Rs)^2	2.29 +/- 1.33 [%]
Orbital Inclination (inc)	83.85 +/- 2.81
Airmass coefficient 1 (a1)	0.58 +/- 0.29
Airmass coefficient 2 (a2)	0.201 +/- 0.041
Scatter in the residuals of the lightcurve fit is	44.11 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.0
Transit Duration (day)	0.071 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.044 vs 0.1214 (deviation 0.51σ)
Duration fit vs published	0.071 d vs 0.0742 d (deviation 0.09σ)
Mid-time O–C	-1.1 min
Depth SNR	2.6
Residual scatter	44.11 %

Light curve

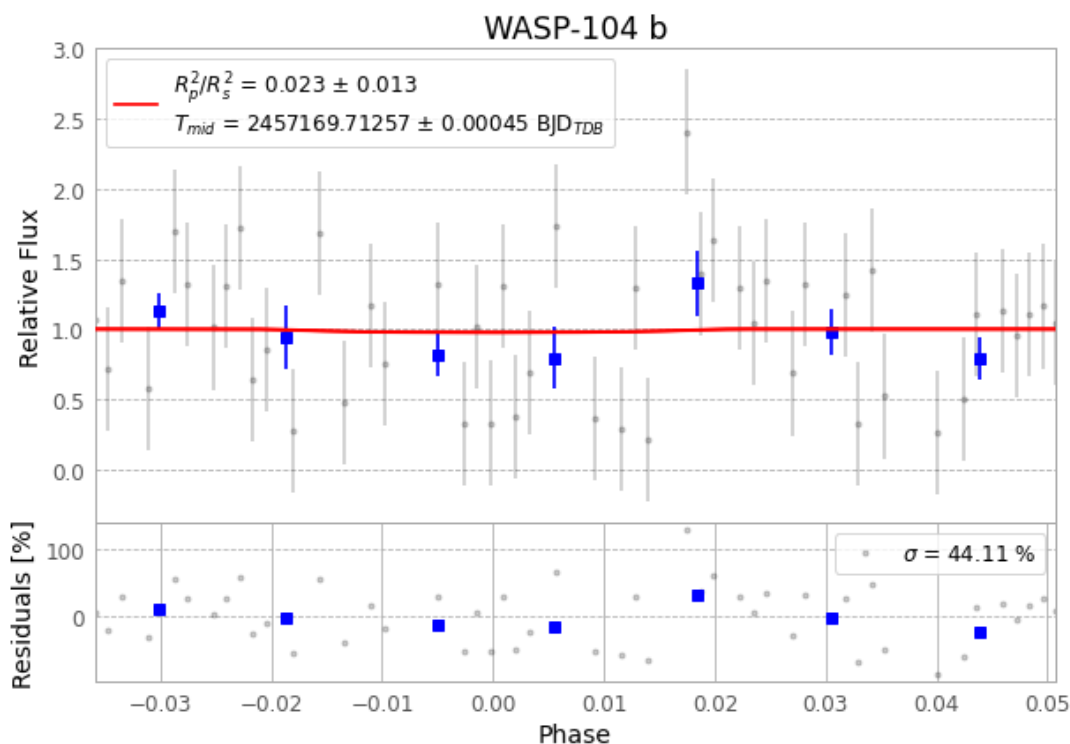


Figure 1 — FinalLightCurve_WASP-104 b_2015-05-26.png (SHA-256 c8ce74814165967a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [505, 92], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e4518161c5d22115...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

79 FITS frames (52 MB), WASP-104150527032412.FITS → WASP-104150527071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150527.log (7a3725049816...), FinalLightCurve_WASP-104 b_2015-05-26.png (c8ce74814165...), FinalLightCurve_WASP-104 b_2015-05-26.csv (b1eade39ddf8...)

Compute resources

Wall time 123.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:30Z.